INCH-POUND

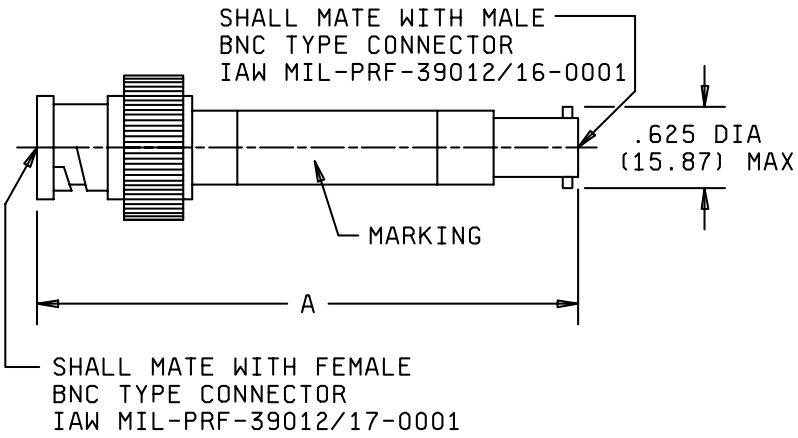
MIL-DTL-3933/19B
w/AMENDMENT 1
8 January 2008
SUPERSEDING
MIL-DTL-3933/19B
7 November 2002

DETAIL SPECIFICATION SHEET

ATTENUATORS, FIXED, COAXIAL LINE (BNC CONNECTORS), FREQUENCY RANGE: DC TO 4.5 GHz, LOW POWER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and the latest issue of MIL-DTL-3933.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 1. Dimensions and configuration.

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ENGINEERING DATA:

Attenuators for use in general field equipment.

Operating frequency range: DC to 4.5 GHz.

Test frequency range for SWR and attenuation measurements, measured or recorded, when using an Automatic Network Analyzer (ANA) or equivalent test equipment: 100 MHz (or lower) to 4.5 GHz.

Basic test frequency for sensitive measurements like connector repeatability, temperature sensitivity, power sensitivity and any other measurements where a 'delta' specification is imposed: 2.0 GHz.

Operating temperature range: -55°C to 125°C.

Maximum weight: See table I.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Nominal impedance: 50 ohms.

Attenuation: See table I. Stability and sensitivity:

Maximum change in attenuation after:	**up to 10 dB, inclusive (dB)**	**over 10 dB (dB/dB)**
Temperature change or thermal shock	0.05	0.005
Vibration or shock	0.1	0.01
Moisture resistance or salt spray	0.2	0.02
Peak power	0.05	0.005

Frequency sensitivity, maximum: 0.1dB/dB/GHz.
Temperature sensitivity of attenuation, maximum: 0.0004 dB/dB/°C.

Power rating: See table I.
Power sensitivity for full input power, maximum: 0.005 dB/dB/Watt.

VSWR: See table I.

Connector repeatability, maximum variation in attenuation: 0.02 dB.

Part or Identifying Number (PIN): M3933/19- (and dash number from table I).

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TABLE I. Electrical characteristics.

Dash number	Attenuation dB		Maximum power input at 25°C		VSWR maximum 1/				Weight (oz) Maximum	Dimension A (inches) Maximum
	Nominal	Deviation DC to 4.5 GHz	AV (W) 2/ (continuously)	Peak (kW)	DC to 1.0 GHz	1.0 to 2.0 GHz	2.0 to 3.0 GHz	3.0 to 4.5 GHz		
01 N S	1	±0.3	4	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
02 N S	2	±0.3	4	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
03 N S	3	±0.3	4	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
04 N S	4	±0.3	3	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
05 N S	5	±0.3	3	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
06 N S	6	±0.3	3	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
07 N S	8	±0.5	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
08 N S	10	±0.5	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
09 N S	11	±0.5	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
10 N S	12	±0.5	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
11 N S	15	±0.5	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
12 N S	20	±0.5	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)

See footnotes at end of table.

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TABLE I. Electrical characteristics - Continued.

Dash number	Attenuation dB		Maximum power input at 25°C		VSWR maximum 1/				Weight (oz) Maximum	Dimension A (inches) Maximum
	Nominal	Deviation DC to 4.5 GHz	AV (W) 2/ (continuously)	Peak (kW)	DC to 1.0 GHz	1.0 to 2.0 GHz	2.0 to 3.0 GHz	3.0 to 4.5 GHz		
13 N S	25	±0.9	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
14 N S	30	±0.9	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
15 N S	40	±0.9	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	3.0 (76.2)
16 N S	50	±1.2	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	3.0 (76.2)
17 N S	60	±1.4	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	3.0 (76.2)
18 N S	13	±0.5	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)

1/ VSWR value is for both ends.

2/ Power input is derated linearly to 0.5 watts at +125°C.

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Referenced documents. In addition to MIL-DTL-3933 this specification sheet references the following documents: MIL-PRF-39012/16 and MIL-PRF-39012/17.

Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5985-2008-13)

Review activities:
Army - MI
Navy - AS, MC, SH
Air Force - 99

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